Clinical Study Summary Report (CSSR)

Document Status: Draft | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: Phase 2 Study Of VLA15, A Vaccine Candidate Against Lyme Borreliosis, In A Healthy Pediatric And Adult Study Population **Short Title/Acronym:** VLA15-221 **Protocol Number:** VLA15-221 **NCT Number:** NCT04801420 **Sponsor Name:** Valneva and Pfizer **Investigational Product:** VLA15 (Multivalent recombinant protein subunit vaccine targeting OspA) **Phase of Development:** Phase 2 **Medical Monitor:** Pfizer/Valneva Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate and compare the safety and immunogenicity of two different primary immunization schedules of the VLA15 vaccine (Month 0-2-6 vs. Month 0-6) in a healthy pediatric, adolescent, and adult population. * **Secondary Objectives:** To assess the safety and immunogenicity (anamnestic antibody response) of annual booster doses administered at Months 18, 30, and 42, and to monitor long-term antibody persistence.

2.2 Background & Rationale Lyme disease is the most common vector-borne illness in the Northern Hemisphere, caused by *Borrelia burgdorferi* and transmitted via tick bites. With rising global incidence and no currently approved human vaccine, VLA15—a novel recombinant vaccine targeting six outer surface protein A (OspA) serotypes—shows promise as an effective preventive strategy. Blocking OspA inhibits the bacterium's ability to leave the tick and infect humans. This study aims to optimize the dosing schedule and extend prophylactic protection to children and adolescents.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled * **Blinding:** Observer-blind * **Randomization:** Randomized (1:1:1 ratio for the respective dose/schedule cohorts)

3.2 Planned Enrollment & Duration * **Target N:** 625 enrolled subjects (initially targeted 600) * **Number of Sites:** 14 clinical study centres in Lyme borreliosis-endemic areas in the USA. * **Estimated Study Start:** March 15, 2021 * **Estimated Primary Completion:** Ongoing (Total maximum study duration per subject is up to 50 months).

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy pediatric, adolescent, and adult subjects aged 5 to 65 years. 2. Subjects with a history of Lyme borreliosis (previous infection with *Borrelia*) as well as *Borrelia* naïve subjects. 3. If of childbearing potential: Negative serum pregnancy test at screening and agreement to employ adequate birth control measures for the required timelines.

4.2 Exclusion Criteria 1. Chronic illness related to Lyme borreliosis (LB), active symptomatic LB, or treatment for LB within 3 months prior to Day 1. 2. Previous vaccination against LB. 3. A tick bite within 4 weeks prior to Day 1. 4. Known thrombocytopenia, bleeding disorder, or receipt of anticoagulants in the 3 weeks prior to Day 1.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Intramuscular injections of 180 µg VLA15 or placebo. Administered under two primary schedules: a 3-dose schedule (Months 0, 2, and 6) or a 2-dose schedule (Months 0 and 6). Booster doses are administered at Months 18, 30, and 42. * **Route of Administration:** Intramuscular injection. * **Duration of Treatment:** Up to 50 months per subject (including the Booster Phase).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard concomitant medications not affecting immune response. * **Prohibited:** Active or passive immunization within 4 weeks prior to any vaccination (except for influenza vaccines); anticoagulants within 3 weeks prior to any booster vaccination.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days/Months)	Key Activities/Procedures
Screening	Day 0	Informed Consent, Medical History, Pregnancy Test, Randomization
Primary Phase	Months 0, 2, 6, 7	Vaccination series (M0-2-6 or M0-6), Safety Assessment, Day 208/Month 7 Immunogenicity Blood Draw
Interim / Booster	Months 12, 18, 19, 30, 31, 42, 43	Annual booster vaccinations (M18, M30, M42), Month 19/31/43 Immunogenicity Assessments, Solicited AE tracking
End of Study	Month 50	Final Vital Signs, Exit Interview, Safety/ Antibody Persistence Check

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** 1. *Immunogenicity:* OspA serotype (ST)-specific IgG geometric mean titers (GMTs) against

ST1 to ST6 at Month 7 (Day 208). 2. *Safety:* Frequency of solicited local and systemic adverse events occurring within 7 days after each and any vaccination. * **Measurement Method:** IgG binding assay (ELISA) and patient symptom diaries.

7.2 Secondary & Exploratory Endpoints * Frequency of solicited local and systemic adverse events within 7 days after each booster dose. * OspA serotype-specific IgG GMTs evaluated one month after each respective booster (i.e., Month 19, Month 31, and Month 43) to assess anamnestic antibody responses. * Long-term antibody persistence evaluated at interim and follow-up visits (e.g., Month 12, Month 50).

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Solicited AEs are collected via patient diaries for 7 days post-vaccination. Unsolicited AEs are tracked up to specified follow-up periods.
- **Serious Adverse Events (SAEs):** Continuously monitored throughout the 50-month study duration.
- **Data Monitoring Committee (DMC):** Regular trial safety data reviews conducted by specific unmasked internal/external personnel and statisticians.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Analysis Set (all individuals who received at least one vaccination); Per-Protocol Analysis Set (participants who received vaccinations and provided valid serological samples without major protocol deviations).
- **Sample Size Justification:** $N=625$, split into an age-descending enrollment process (18-65 years, 12-17 years, and 5-11 years in a 2:1:1 ratio) to ensure robust statistical powering across all age demographics.
- **Handling of Missing Data:** Standard imputation methodology applied for IgG GMT analyses per the SAP.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Continuous site monitoring; unmasking limited to site staff handling the investigational product and specific safety statisticians.
- **Audit Trail:** Utilizing an electronic data capture system for randomization, patient diaries, and eCRF locks.

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11. Study Identifiers & Governance

- **Primary ID:** VLA15-221
- **Registry Number:** NCT04801420
- **Sponsor/Lead Organization:** Valneva and Pfizer
- **Study Title:** Phase 2 Study Of VLA15, A Vaccine Candidate Against Lyme Borreliosis, In A Healthy Pediatric And Adult Study Population
- **Official Regulatory Title:** Phase 2 Study Of VLA15, A Vaccine Candidate Against Lyme Borreliosis, In A Healthy Pediatric And Adult Study Population

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12. Clinical Framework (Lyme Borreliosis Specific)

- **Primary Condition:** Lyme Borreliosis (Lyme Disease)
- **Diagnosis Threshold:** N/A (Healthy populations and individuals with past exposure included for prophylactic vaccination)
- **Medical Methodology:** Recombinant multivalent protein subunit prophylactic vaccine
- **Optimization Target:** Seroconversion (above 90%) and anamnestic antibody response across all 6 OspA serotypes

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13. Study Procedures & Methodology

- **Screening & Baseline:** Informed consent acquisition, medical history review, pregnancy testing, and baseline randomization procedures.
- **Management Pathway:** Scheduled vaccination intervals optimized for tick season prophylactic readiness.
- **Education Protocol (HCP):** Blinded administration guidelines, reporting parameters for local/systemic reactions.
- **Education Protocol (Patient):** Self-reporting of solicited adverse events (fever, headache, fatigue, arthralgia) via daily logs post-vaccination.
- **Quality Audit Mechanism:** Periodic masked safety data reviews by the sponsor to assess vaccine tolerability before proceeding to pediatric dose expansions.
- **End of Study Closeout:** Final vital signs and exit interview conducted at Month 50 to assess final safety and antibody persistence.

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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 50 months
- **Onsite Visit Cadence:** Months 0, 2, 6, 7, 12, 18, 19, 30, 31, 42, 43, and 50
- **Remote Monitoring Frequency:** Regular interval symptom diary checks (7 days post-dosing).

- **Checklist Review Interval:** Continuous safety monitoring updates evaluated by trial statisticians.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]